

CS Life

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Team Name: BitByte

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CS Life

- Players experience the life of a CS student over four years of college.
- The game progresses one academic quarter at a time.
- Players must balance important aspects of college life, including:
 - Studying and academics
 - Social life and friendships
 - Relationships
 - Finances and budgeting
 - Personal health and well-being
 - Career preparation and skill development
- Each quarter, players make decisions that affect their character's stats
- Different choices lead to different experiences and outcomes
- By graduation, players may achieve various endings based on their stats



Project Scope

1. Develop a fully playable CS college life simulator in which players manage a student's college experience.
2. Implement a character quarterly progression system with basic statics and player decision options
3. Create the ability to allow for one complete playthrough from the beginning of freshman year to graduation with multiple possible endings
4. Generate random events that could occur at any moment and can drastically change the outcomes.

Create Your Character

Name

Trait

Start Game



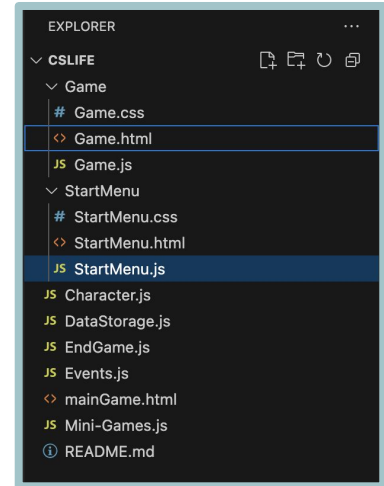
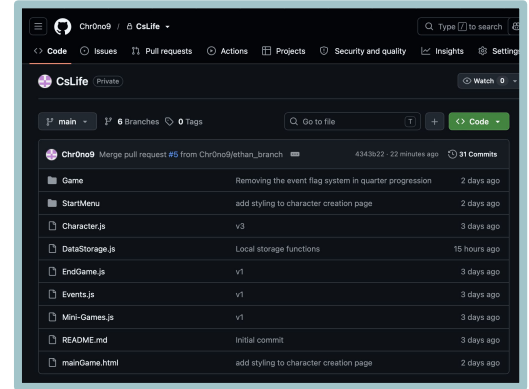
Sprint Plan 1

- User Stories:
 - As a player, I want to start a new game in which I can choose my name, avatar and starting stats (Goal 1)
 - As a player, I want the ability to view my character's stats (Goal 2)
 - As a player I want to be able to progress through quarters (Goal 2)
 - As a user, I want my progression to be saved (Goal 3)
- Spikes:
 - Research how to store and retrieve data (name, stats, current quarter) using Local Storage.
 - Find different ways to display player stats through a modal or pop-up window.
 - Research how to implement a quarter progression system that updates the player's current quarter and game state.
 - Research different methods for implementing different avatars.



Sprint Plan 1

- Infrastructure Tasks:
 - Setting up a Git Repository
 - Set up separate branches for all team members
 - Separate Game and Starting Menu folders with their own HTML, CSS and JavaScript files
 - Create a save/load system using local storage



Sprint Plan 2

- User Stories:
 - As a player, I want the ability to choose how I spend my time each quarter
 - As a player, I want to visualize how choices affect my stats and change outcomes
 - As a user, I want random life events a real CS student might experience to occur.
- Spikes:
 - Research how to generate random life events using JavaScript.
 - Investigate how to modify player stats dynamically based on user choices.
 - Research how to build a random life events system and choices system that will trigger between quarters.
- Infrastructure Tasks:
 - Create a random event data structure to store event names, descriptions, and stat effects.
 - Create a JavaScript function to update player stats based on the player's choices.



Sprint Plan 3

- User Stories:
 - As a player interested in CS, I want to improve my programming skills
 - As a player interested in CS, I want to prep for job interviews
 - As a player interested in CS, I want to visualize how different choices can bring forth opportunities.
- Spikes:
 - Research and prototype simple coding challenges (i.e. multiple choice, short debugging questions, leetcode questions)
 - Research common interview topics and how they could be represented as game activities.
 - Investigate systems for tracking project completion across multiple quarters.
- Infrastructure Tasks:
 - Build a reusable activity selection system for CS related actions.
 - Implement functions to update player stats after completing CS activity
 - Create data structures for coding challenges and interview prep activities



Sprint Plan 4

- User Stories:
 - As a player, I want graduation outcomes to differ based on my choices, jobs, and stats
 - As a player, I want to see how my choices lead to a certain graduation outcome
 - As a player, I want to see how my decisions over four years shaped my academic, social, and professional success.
- Spikes:
 - Investigate how to determine outcomes while taking all stats into consideration.
 - Research different approaches for creating multiple unique endings and career paths.
 - Look over the product for potential improvements and changes.
- Infrastructure Tasks:
 - Build the graduation outcome framework that evaluates player statistics and determines final outcome
 - Implement a statistics summary screen showing final academic, social, and career performance.



Architecture

Frontend (UZ)

- 1) character creation screen - select appearance, name, and initial stats
- 2) statistics interface - view stats of character
- 3) mini-games - do coding challenges and play mini-games
- 4) career progression - internships and job interviews based on stats.
- 5) quarter progression - keep track of current quarter and go to next one.
- 6) outcomes - view graduation outcome based on your stats.

Core Game Systems

1. Character Creation

- choose appearance
- choose name
- set initial stats/traits

2. Stats System

- 1) health
- 2) happiness
- 3) money
- 4) relations
- 5) GPA / intellect
- 6) coding skills
- 7) career readiness

3. Quarterly Progression

- 4 years / 16 quarters
- decisions affect stats for each quarter.
- time moves the story forward

4. Decisions

- Player allocates time across activities.
- leads to mini-games.
- impacts stats/outcomes.

5. random events

- 1) car problems
- 2) burnout / illness
- 3) job offers
- 4) academic issues
- 5) relationship issues
- 6) and more...

6. Mini-games

- 1) coding challenges
- 2) study sessions
- 3) internship tasks
- 4) social / relationship scenarios.
- 5) interviews.
- 6) and more...

7. Outcomes

- evaluate stats to determine:
- 1) jobs
 - 2) internships
 - 3) dropout
 - 4) relationships
 - 5) grad school
 - 6) unemployed.

Data Storage (local / browser)

Stores:

- Player profile
- stats
- game progress
- quarter
- decision history

Tech-Stack

Frontend:

- HTML / React
- CSS
- Javascript

Data Storage:

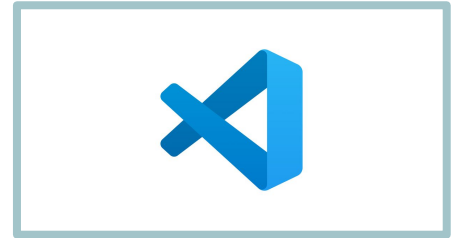
- Local storage
- API

Other tools:

- Github
- VS code

Technologies

- Frontend
 - HTML/React
 - CSS
 - Javascript
- Data Storage
 - Local Storage API
- Other Tools
 - Github
 - VS code



Challenges/Risks

- Large Feature scope for a short timeline
 - Character creation, stats, random events, quarters, jobs, saving/loading, internships/jobs
- Keeping game systems connected
 - Stats, choices, random events...
- Balancing the gameplay
 - Choices should feel meaningful without making the game too easy or punishing
- Save/load complexity
 - As more features are added, local storage will have to save all important game state correctly
- Merge Conflicts or file structure changes
 - Teammates may update files at the same time



Minimal Viable Product(MVP)

- Playable browser game in which the user can successfully create a character with distinct starting statistics
- The player should be able to track their progress and development through academic quarters reaching all the way to graduation.
- The player player will have to make decisions about how to dedicate their time, energy, and money towards different things such as classes, social life, coding skills, and more.
- The user will experience random life events in which they will select life altering choices.
- The project is a complete success if the user can complete one full playthrough from character creation to the end of college with a proper graduation feature giving them different outcomes based on how they did throughout their journey.
- The game should have basic save/load functionality.



Minimal Viable Product(MVP)

User Stories

- As a player, I want to create a character by choosing a name and starting trait.
- As a player, I want the ability to view my character's stats.
- As a player, I want to progress through academic quarters.
- As a player, I want to choose how I dedicate my time each quarter.
- As a player, I want my choices to affect my character's stats.
- As a player, I want random life events to occur throughout the game.
- As a player, I want different graduation outcomes based on my choices and overall performance.



Thank You!



Any Questions/Feedback?